JAN-M-560

12 FEBRUARY 1948

JOINT ARMY-NAVY SPECIFICATION

MAGNESIUM STEARATE (OINTMENT TYPE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective departments and supersedes the following specification:

Chemical Corps
Tentative Specification
197-54-354
11 Feb. 45

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN-P-118—Packaging and Packing for Overseas Shipment—Sacks, Paper, Shipping, Multiwall.

U. S. ARMY SPECIFICATION

100-2—Standard Specifications for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATIONS

P-S-536—Soap and Soap-Products; General Specifications (Methods for Sampling and Testing).

RR-S-366—Sieves; Standard, Testing.

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

B. GRADE

B-1. This specification covers one grade of magnesium stearate.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—The magnesium stearate shall be manufactured from a high grade stearic acid.

D. GENERAL REQUIREMENTS

D-1. See section E.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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E. DETAIL REQUIREMENTS

E-1. The material shall comply with the detail requirements listed in table I.

Table I.—*Detail requirements.*

	Minimum	Maximum
Particle size (retained by No. 325 U. S. Standard sieve[1]) .....(percentage)	....	0.50
Magnesium oxide (dry basis)..... ..... (percentage)	7.40	8.00
Extraneous ash (dry basis) (percentage)	....	0.80
Moisture	....	4.00
Acetone-soluble matter ....(percentage)	....	2.00
Iodine number of separated fatty acid...	....	7
Grit	....	none
Trace elements (calculated as metal)[2]..		
Lead .....(ppm.)	....	25
Arsenic .....(ppm.)	....	10
Copper .....(ppm.)	....	15
Iron .....(ppm.)	....	100

[1] Sieves shall conform to Federal Specification RR-S-366.
[2] The manufacturer shall, at the time of forwarding his bid, submit for approval by the procuring agency, a complete description of the methods of analysis he proposes to use in the certification required in paragraph F-4h.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F-1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F-2. *Size of lot.*—Unless otherwise specified in the contract or order, each batch shall constitute a lot. (See par. H-5.)

F-3. *Sampling.*—Five containers shall be chosen, at random, from each lot for sampling purposes. If there are fewer than five containers, sample each container. Withdraw a ½-pound representative sample from each container so chosen. Place the samples in a clean, dry container and mix thoroughly to form a composite sample representative of the lot. Seal and label the container to identify the sample with the lot represented. (See par. H-6.)

F-4. *Tests.*—All tests should be conducted in accordance with the procedure stated herein. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents as used in the tests, and corrections shall be applied when significant.

F-4a. *Particle size.*—Weigh approximately 5.00 gm. of sample into a beaker. Add 100 ml. denatured ethyl alcohol and stir vigorously. Immediately pour the suspension into a No. 325 U. S. Standard sieve. Wash the residue with alcohol and brush the residue gently through the screen with a camel hair brush. (See par. H-3.) Wash the brush off with alcohol and dry the residue on the screen in an oven at 100° to 110°C. Cool in a desiccator and weigh.

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$$\text{Percentage retained by a No. 325 sieve} = \frac{\text{weight of residue} \times 100}{\text{weight of sample}}$$

F–4b. *Magnesium oxide.*—Weigh approximately 1.500 gm. of sample into a tared platinum crucible. Ignite to constant weight, cool in an efficient desiccator and weigh the ash, which is very hygroscopic, as rapidly as possible. (See par. F–4c.) Treat the ash with 5 ml. of 1:1 hydochloric acid and 20 ml. of water. Heat to effect solution and filter into a 250-ml. beaker. Wash the filter free of chlorides with hot water, add 5 ml. of concentrated hydrochloric acid, dilute to 125 ml., add 1.2 gm. of ammonium diphosphate. Add 5 gm. of ammonium acetate and a few drops of phenolphthalein solution. Heat to near boiling, then slowly add 1.5*M* ammonium hydroxide until a faint pink color is obtained and slight precipitation occurs. Stir well for 1 minute, touching the sides of the beaker as little as possible. When the precipitate has become distinctly crystalline, add more ammonium hydroxide until a deep red color is obtained. Allow the solution to cool and add a quantity of concentrated ammonium hydroxide equal to one-fifth of the volume of the solution. Allow to stand for at least 4 hours, preferably over night. Filter through a tared Gooch crucible. Wash the precipitate with 1.5*M* ammonium hydroxide until free of chlorides and finally moisten with 4 or 5 drops of a saturated solution of ammonium nitrate in 1.5*M* ammonium hydroxide. Ignite very slowly, gradually increasing the heat until the precipitate is white. Cool in a desiccator and weigh as magnesium pyrophosphate.

$$\text{Percentage MgO on dry basis} = \frac{\text{weight of } Mg_2P_2O_7 \times 36.23 \times 100}{\text{weight of sample} \times (100 - \text{percentage moisture})}$$

F–4c. *Extraneous ash.*—Calculate the percentage extraneous ash, from the weight of ash found as described in paragraph F–4b, as follows:

$$\text{Percentage extraneous ash on dry basis} = \frac{\text{weight of ash} \times 100 \times 100}{\text{weight of sample} \times (100 - \text{percentage moisture})} - \text{percentage MgO}$$

F–4d. *Moisture.*—Determine the moisture content of the material by the oven method prescribed in test No. 20.1 of Federal Specification P–S–536.

F–4e. *Acetone-soluble matter.*—Weigh approximately 1.50 gm. of sample into a 250-ml. beaker and add 50 ml. of hot acetone. Cover the beaker with a watch glass and heat carefully on a steam bath for 5 minutes. (Warning: Keep away from flame.) Filter into a tared evaporating dish and wash the residue with five 10-ml. portions of boiling acetone. Evaporate filtrate and washings on a steam bath, dry to constant weight in an oven at 105° to 110°C., cool in a desiccator, and weigh.

$$\text{Percentage acetone-soluble matter} = \frac{\text{weight of residue} \times 100}{\text{weight of sample}}$$

F–4f. *Iodine number.*—

F–4f(1). *Separation of fatty acid.*—Place about 50 gm. of sample in a large porcelain dish and decompose by heating with an excess of diluted hydrochloric acid. Continue the decomposition until the fatty layer covering the surface of the liquid is melted and clear in appearance. Cool and remove the solidified fatty acid. Purify the fatty acid by boiling four times with water, cooling the fatty acid and removing it to new portions of water each time. Dry the solid acid between filter papers, break into small pieces, place on a rapid filter paper which drains into a beaker. Place the assembly in an oven at approximately 100°C., where the acid will melt and filter into the beaker. The resulting fatty acid should be clear and water-free.

F–4f(2). *Iodine number.*—Determine the iodine number of the separated fatty acid (see par. F–4f(1)) in accordance with the method prescribed in Test No. 110.1 of Federal Specification P–S–536.

F–4g. *Grit.*—Place a quantity of the sample on a smooth glass plate and pulverize with a smooth steel blade. Consider as grit any rough foreign particles which tend to give the characteristic tactile sensation experienced when sand is ground between steel and glass in the same manner.

F–4h. *Trace elements.*—The manufacturer shall certify that the trace elements are present in proportions not greater than those listed as maxima in table I.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—Unless otherwise specified in the contract or order, the material shall be packed in 50-pound capacity, 5-ply multiwall paper shipping sacks complying with style 2–K of Joint Army-Navy Specification JAN–P–118. Types C and D sacks, when supplied, shall be constructed with tuck-in-sleeves of class a, grade I, creped paper.

G–2. *Marking.*—Each container shall be clearly marked "Suitable for use in Ointments", together with the date of manufacture and maximum storage temperature. In addition to the above and any special marking required by the contract or order, shipping containers for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Material covered by this specification is intended for use as a component of protective ointment.

H–2. Requests, requisitions, schedules, and contracts or orders should specify the title, number, and date of the specification.

H–3. Alcohol, used in sieve size determination, may be filtered and reused.

H–4. Material complying with the extremes of the limit specified in table I, for magnesium oxide, extraneous ash, and acetone-soluble matter, has been used with acceptable results. However, it was found that best results were obtained when the magnesium

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oxide approached the upper limit and the extraneous ash and acetone-soluble matter approached zero.

H–5. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–6. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion and with the concurrence of the contractor, may modify, in whole or in part, the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department". Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Department of the Army and the Department of the Air Force will obtain copies of Joint Army-Navy, U. S. Army and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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[JAN-M-560]

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